



Science in Ancient and Medieval India

Mathematicians and Astronomers of Ancient India – Aryabhata

Aryabhata was one of the **greatest mathematicians and astronomers of ancient India**. He made pioneering contributions to **mathematics, algebra, trigonometry, and astronomy** during the Gupta period, often called the Golden Age of Indian science.

- **Born:** 476 CE
- **Place:** Traditionally associated with Kusumapura (modern Patna)
- **Period:** Gupta Age

He is considered the **founder of classical Indian astronomy**.



Major Work

Aryabhatiya

- Written in 499 CE.
- Composed in Sanskrit verse.
- Contains 121 shlokas.
- Divided into four sections:
 1. Gitikapada
 2. Ganitapada
 3. Kalakriyapada
 4. Golapada

This text summarizes his discoveries in mathematics and astronomy.



Contributions to Mathematics



1 Concept of Zero and Place Value

- Worked with the decimal place value system.
- Used zero as a placeholder (though symbol developed later).
- Strengthened the base-10 number system.



2 Value of Pi (π)

- Calculated π as **3.1416** (very accurate for that time).
- Stated that π is an approximation.

3 Algebra

- Solved linear equations.
- Used algebraic methods to solve indeterminate equations (Kuttaka method).

4 Trigonometry

- Introduced sine (jya) and cosine functions.
 - Prepared a table of sine values.
 - Developed trigonometric formulas.
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Contributions to Astronomy

1 Rotation of the Earth

- Stated that the Earth rotates on its axis.
- Explained day and night due to Earth's rotation.
- This was revolutionary at that time.

2 Solar and Lunar Eclipses

- Correctly explained eclipses scientifically.
- Rejected mythological explanations.
- Said eclipses occur due to shadow of Earth and Moon.

3 Length of the Year

- Calculated the length of the solar year as:
365 days, 6 hours, 12 minutes, 30 seconds
- Very close to modern value.

4 Planetary Motion

- Studied motion of planets.
 - Calculated positions of celestial bodies.
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Scientific Methods

- Used observation and calculation.
 - Combined mathematics with astronomy.
 - Developed astronomical instruments and computational techniques.
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Importance in Ancient Indian Science

- Laid foundation for later scholars like:
 - Brahmagupta
 - Bhaskara I
 - Bhaskara II
 - Influenced Islamic and European astronomy through translations.
 - His works were translated into Arabic during the Abbasid period.
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Scientific Outlook

Aryabhata showed:

- Rational thinking
- Mathematical precision
- Rejection of superstition
- Scientific explanation of natural phenomena

He represents the advanced level of scientific development in ancient India.



Conclusion

Aryabhata was a pioneering mathematician and astronomer whose work in the **Aryabhatiya** revolutionized ancient science. His accurate calculations of π , explanation of eclipses, theory of Earth's rotation, and development of trigonometry place him among the greatest scientists of the ancient world.

He stands as a symbol of the rich scientific tradition of **Ancient India**.



Science in Ancient and Medieval India

Mathematicians and Astronomers of Ancient India – Varahamihira

Varahamihira was one of the **most eminent scholars of the classical period of Indian astronomy and mathematics**, during the **Gupta Age (6th century CE)**. He was a polymath who contributed to **astronomy, astrology, mathematics, geography, and meteorology**.

He is considered one of the “**Nine Gems (Navaratnas)**” of King Vikramaditya’s court in Ujjain.



Major Work



Brihat Samhita

- Written around **6th century CE**.
- Encyclopedic work on **astronomy, astrology, architecture, and natural phenomena**.
- Covers topics such as:
 - Planetary positions and eclipses
 - Rainfall, clouds, and meteorology
 - Agriculture and horticulture
 - Architecture (Vastu Shastra) and construction
 - Omens, prognostication, and astrology

◆ Importance:

- Reference for astronomers, astrologers, architects, and meteorologists.
 - Combines observation with practical application.
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Pancha Siddhantika

- Summary of **five earlier astronomical schools (Siddhantas)**:
 1. Surya Siddhanta
 2. Romaka Siddhanta
 3. Paulisha Siddhanta

4. Vasishtha Siddhanta
 5. Paitamaha Siddhanta
- Introduced **methods for calculating planetary positions, eclipses, and trigonometric tables.**

◆ Importance:

- One of the most authoritative classical Indian texts on **astronomical calculations.**
 - Influenced later scholars such as Brahmagupta and Bhaskara I.
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Contributions to Mathematics

1. **Trigonometry**
 - Used sine tables (jya) for astronomical calculations.
 - Introduced relationships between chords and arcs.
 2. **Geometry**
 - Applied geometry in architecture, construction, and astronomy.
 3. **Arithmetic**
 - Developed methods for **almanac calculations** and planetary computations.
 4. **Calendar Calculations**
 - Explained **luni-solar calendar systems** and intercalation of months.
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Contributions to Astronomy

1. **Planetary motion**
 - Calculated planetary positions using Siddhanta methods.
 2. **Eclipses**
 - Explained solar and lunar eclipses scientifically.
 3. **Earth and Celestial Phenomena**
 - Observed solstices, equinoxes, and planetary conjunctions.
 4. **Meteorology**
 - Studied clouds, rainfall, monsoon patterns, and natural signs.
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Other Scientific Contributions

- Astrology: Classified zodiac signs, horoscopes, and omens.
- Architecture: Brihat Samhita provides guidelines for construction, city planning, and iconography.
- Agriculture: Advises on crop cycles based on astronomy.
- Natural Science: Notes on animals, plants, rivers, and geography.

Scientific Outlook

Varahamihira combined **empirical observation with classical knowledge**, integrating:

- Astronomy & astrology
- Mathematics & geometry
- Natural science & environment
- Practical applications for daily life

He is known for a **rational yet holistic approach** to science.

Importance in Ancient Indian Science

- Key figure of **Gupta-era scientific tradition**.
 - Systematized astronomical knowledge of earlier schools.
 - Bridged **mathematics and astronomy with practical applications**.
 - His works influenced **Indian and Islamic astronomy**.
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Conclusion

Varahamihira was a polymath whose works, **Brihat Samhita** and **Pancha Siddhantika**, established him as one of the most important **astronomers, mathematicians, and scientists** of ancient India.

His integration of observation, calculation, and practical application makes him a **pioneering figure in classical Indian science**.



Science in Ancient and Medieval India

Mathematicians and Astronomers of Ancient India – Brahmagupta

Brahmagupta was one of the **greatest mathematicians and astronomers of ancient India**. He made pioneering contributions to **arithmetic, algebra, geometry, and astronomy** and is especially famous for his work on **zero and negative numbers**.

- **Born:** 598 CE, in Bhinmal (Rajasthan)
 - **Period:** 7th century CE, after Aryabhata
 - **Significance:** Built upon Aryabhata's work and advanced Indian mathematics and astronomy significantly.
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Major Work

Brahmasphutasiddhanta

- Written in **628 CE**.
- Composed in **Sanskrit verse**.
- Divided into 25 chapters covering **arithmetic, algebra, geometry, and astronomy**.

◆ Sections Include:

1. **Arithmetic** – Operations with integers, fractions, zero
 2. **Algebra** – Quadratic, indeterminate equations
 3. **Geometry** – Measurement of areas and volumes
 4. **Astronomy** – Planetary motions, eclipses, calendar calculations
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Contributions to Mathematics



Zero and Negative Numbers

- Defined **zero (shunya)** as a number.
- Explained operations with zero: addition, subtraction, multiplication, division.
- Handled **negative numbers (debt)** in calculations.

Example from Brahmagupta:

- “ $a + 0 = a$, $a - 0 = a$, $a \times 0 = 0$, $a \div 0 = a / 0$ ” (rules for operations)
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2 Arithmetic

- Operations on positive and negative numbers.
 - Fractions and decimal calculations.
 - Developed rules for solving linear and quadratic equations.
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3 Algebra

- Solved **indeterminate equations** ($ax + by = c$) using the **Kuttaka method**.
 - Studied quadratic equations and their solutions.
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4 Geometry

- Formula for the area of a triangle using **Heron's formula**.
 - Measurement of circular and square areas.
 - Volumes of pyramids, spheres, and cylinders.
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Contributions to Astronomy

1 Planetary Motion

- Calculated **mean longitudes of planets**.
- Provided accurate methods for astronomical calculations.

2 Eclipses

- Explained **solar and lunar eclipses** mathematically.
- Used trigonometry for prediction.

3 Calendar and Time

- Computed **sidereal year**: 365 days, 6 hours, 12 minutes.
- Discussed intercalary months for lunar calendar.

◆ Scientific and Mathematical Innovations

1. **Systematic treatment of zero and negative numbers**
 2. **Rules for algebraic equations**
 3. **Trigonometry and sine tables** for astronomy
 4. **Practical arithmetic for trade and astronomy**
 5. Integration of mathematics with **astronomy and astrology**
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◆ Importance in Ancient Indian Science

- Extended Aryabhata's work on astronomy and mathematics.
 - First to give **rules for operations with zero and negative numbers**.
 - Influenced later mathematicians:
 - Bhaskara I
 - Bhaskara II
 - His methods reached the **Islamic world** and later Europe.
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◆ Comparison with Aryabhata (Summary)

Feature	Aryabhata	Brahmagupta
Concept of zero	Placeholder only	Number with rules
Negative numbers	Not defined	Defined and used
Algebra	Linear and some indeterminate equations	Quadratic and indeterminate equations (Kuttaka method)
Astronomy	Earth rotation, eclipses, planetary motion	Improved calculations, trigonometry, eclipses
Pi (π)	Approx. 3.1416	Used more precise computations

★ Conclusion

Brahmagupta was a **pioneering mathematician and astronomer** who developed rules for zero, negative numbers, algebra, and astronomy.

His **Brahmasphutasiddhanta** remains a landmark in Indian mathematics and influenced later generations of mathematicians and astronomers globally.
